

Student Placement Prediction

Machine Learning Pipeline & Deployment Analysis

Project Domain: Educational Data Mining

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Algorithm: Decision Tree Classifier

Stack: Python, Scikit-Learn, Streamlit

An analytical study on predicting campus recruitment outcomes using academic performance indicators, soft skills, and project experience.

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1 Project Introduction

The **Placement Predictor** is a decision-support tool designed for students and educational institutions to evaluate campus recruitment readiness. By analyzing variables such as CGPA, IQ, and soft skills, the system provides a binary classification (Placed/Not Placed) to help candidates identify areas of improvement before the actual placement season.

2 Data Pipeline & Preprocessing

The model was developed using a dataset of 10,000 student records. The preprocessing phase ensured data integrity through the following steps:

- ✓ **Exploratory Analysis:** Identification of feature distributions (e.g., IQ range from 41 to 150).
- ✓ **Feature Scaling:** Application of `StandardScaler` to normalize numerical features including IQ, CGPA, and Academic Performance.
- ✓ **Binary Transformation:** Converting Internship Experience and Placement Status into boolean integers (0 and 1).
- ✓ **Class Balancing Check:** Analyzing the split between placed (3,964) and unplaced (6,036) candidates.

3 Technical Feature Set

The predictive engine processes eight critical dimensions of a student’s profile:

Feature	Description	Type
CGPA	Cumulative Grade Point Average	Numerical
IQ	Cognitive ability score	Numerical
Academic Performance	Standardized academic rating	Numerical
Projects	Total technical projects completed	Numerical
Internships	Industry exposure (Yes/No)	Binary
Soft Skills	Communication	Categorical

Table 1: Primary Model Predictors

4 Modeling Methodology

4.1 Decision Tree Classifier

The core logic resides in a **Decision Tree Classifier**. This model was chosen for its interpretability and ability to handle non-linear relationships between academic scores and placement likelihood without requiring significant data transformation.

4.2 Model Evaluation

Performance metrics gathered during the validation phase:

- **Accuracy:** High classification accuracy on balanced test sets.
- **Generalization:** The model avoids overfitting through pruning and depth control.

5 Deployment & Interface Design

The system is deployed as a single-page web application using **Streamlit**.

Application Features

- **Interactive Input:** Real-time sliders for CGPA (1.0–10.0) and IQ (40–150).
- **Aesthetic UI:** Custom CSS injection for a professional “Dark Mode” background with card-style input containers.
- **Real-time Feedback:** Utilization of Streamlit balloons and success components for positive placement predictions.

6 Inference Logic

To maintain consistency between the training environment and production, the application employs a serialized pipeline:

1. Loads the pre-trained `DecisionTreeClassifier` and `StandardScaler` via `joblib`.
2. Constructs a 1-row `DataFrame` matching the training schema.
3. Applies the same scaler transform used on the training set.
4. Outputs a binary prediction reflected in the UI.

7 Conclusion

This project successfully bridges data analysis and practical application. By leveraging a Decision Tree model, we’ve created a transparent tool that not only predicts outcomes but also highlights the importance of internships and academic consistency in the recruitment process.